Deformation of the Japanese Islands as inferred from Rock Magnetism

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Summary

Results are presented of the directions of magnetization for twenty-eight sets of rocks collected from different parts of Japan. The directions for rocks of Pre-Tertiary age from south-west Japan differ systematically from those from north-east Japan. It is suggested that the difference is due to the deformation of the Japanese Islands in the late Mesozoic or early Tertiary. Because of the deformation, it is not possible to infer the accurate position of the geomagnetic dipole with respect to Japan for Pre-Tertiary times.

1. Introduction

Oriented rock samples have been collected over a wide area of the Japanese Islands of Hokkaido, Honshu, Shikoku and Kyushu. The collecting sites are

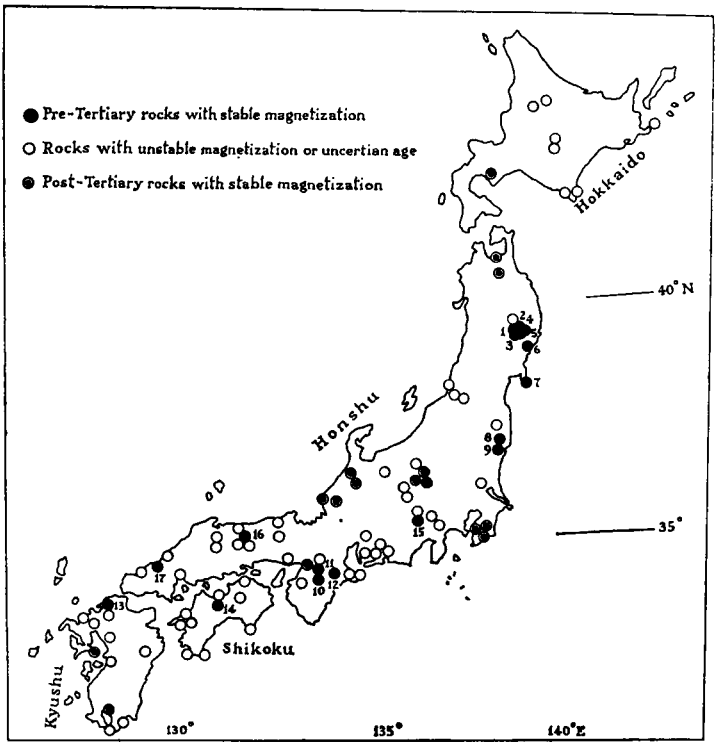


FIG. 1.—The location of sites where rocks were collected. The numbered sites are for rocks of Pre-Tertiary age and the numbers correspond to those given in Table 1 and Figures 2 and 3.

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Table 1

N.R.M. of Pre-Tertiary rocks

	Locality	Rock type	Geological period	D	I	S	α	K	J_r	J_i
			From north-east parts of Japan							
1.	Miyamori (Iwate)	Serpentine	Mesozoic	N 4 W	+51	7	9·3	43·4	87·30	24·70
2.	Tono (Iwate)	Gabbro	Mesozoic	N 106 W	+58	9	6·9	57·3	50·30	79·30
3.	Tono (Iwate)	Serpentine	Mesozoic	N 28 W	+57	5	16·8	21·7	220·00	35·90
4.	Hitokabe (Iwate)	Granite	Mesozoic	N 32 W	+74	10	16·2	9·9	3·14	1·86
5.	Yamaya (Iwate)	Granite	Mesozoic	N 28 W	+53	12	7·7	33·1	11·20	9·89
6.	Takata (Iwate)	Grano-diorite	Mesozoic	N 1 W	+56	12	9·1	23·5	12·60	13·72
7.	Kinkazan (Miyagi)	Granite	Mesozoic	N 57 E	+53	10	9·0	29·6	8·49	11·40
8.	Tono (Fukushima)	Gabbro	Mesozoic	N 53 W	+38	10	10·2	23·3	1·75	0·17
9.	Nakoso (Fukushima)	Gabbro	Mesozoic	N 53 W	+51	12	6·5	45·0	50·30	14·60
			From south-west parts of Japan							
10.	Kohse (Nara)	Serpentine	Late Palaeozoic or Early Mesozoic	N 74 E	+35	7	3·7	265·5	—	—
11.	Sakurai (Nara)	Gabbro	Late Palaeozoic or Early Mesozoic	N 52 E	+43	7	11·7	27·6	5·97	0·44
12.	Taki (Mie)	Gabbro	Late Palaeozoic or Early Mesozoic	N 9 E	+56	5	4·4	306·7	4·21	2·61
13.	Sasakuri (Fukuoka)	Serpentine	Mesozoic	N 29 E	+55	9	17·4	13·0	13·70	6·38
14.	Niihama (Ehime)	Serpentine	Late Palaeozoic or Early Mesozoic	N 13 E	+40	10	18·3	7·9	94·30	66·20
16.	Ishigoshi (Okayama)	Granite	Mesozoic	N 6 W	+42	10	7·1	47·8	3·45	4·50
17.	Nijo (Shimane)	Grano-diorite	Mesozoic	N 34 E	+38	15	14·5	7·9	4·61	10·41

D = Declination. I = Inclination (deg). S = Number of rock samples. α = Radius of 95 per cent circle of confidence (deg) (Fisher 1953). K = Fisher's precision factor. J_r = intensity of remanent magnetization, J_i = intensity of induced magnetization(10^{-5} emu/g).

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shown in Figure 1. The rocks range in age from late Palaeozoic to Quaternary and include a variety of rock types.

2. Directions of natural remanent magnetization

Laboratory measurements of natural remanent magnetization (n.r.m.) are summarized in Tables 1 and 2. The method of measurement and accuracy to be expected for individual measurements have been given by Kawai (1951). For the interpretation given later, it is convenient to group the results from north-east and south-west Japan separately. The directions of n.r.m. for different sites are shown on Schmidt's equal area projections in Figures 2 to 5. Figures 2 and 3 show the directions for Pre-Tertiary rocks from north-east and south-west Japan respectively and Figures 4 and 5 those for Tertiary and Quaternary rocks from north-east and south-west Japan.

Table 2

N.R.M. of Tertiary and Quaternary Rocks

	Locality	Rock type	D	I	S	α	K
		From north-east parts of Japan					
18.	Asama (Nagano)	Andesite	N 5°E	+37°	17	8·1°	20·6
19.	Asama (Nagano)	Andesite	N 10E	+49	24	4·9	37·3
20.	Showashinzan (Hokkaido)	Basalt	N 10E	+57	4	8·5	117·6
21.	Arafune (Nagano)	Porphyrite	N 24E	+49	8	3·9	205·9
22.	Towada (Aomori)	Andesite	N 0	+56	18	4·8	—
		From south-west parts of Japan					
23.	Tomuro (Ishikawa)	Basalt	N 1E	+48	22	9·2	12·3
24.	Tomoru (Ishikawa)	Andesite	N 7W	+50	12	7·2	37·8
25.	Shigisan (Nara)	Andesite	N 4W	+59	20	3·5	84·3
26.	Sakurajima (Kagoshima)	Andesite	N 7E	+45	49	5·5	14·8
27.	Tojimbo (Fukui)	Andesite	N 18E	+54	44	15·9	19·5
28.	Sugiyama (Fukui)	Andesite	N 39E	+49	10	6·4	57·5
29.	Ibaragi (Osaka)	Tuff	3 6W	−51	15	5·4	58·6
30.	Genbudo (Hyogo)	Basalt	S 13W	−35	10	6·1	63·8

D = Mean declination of n.r.m.
I = Mean inclination of n.r.m. (deg).
S = Number of rock samples.
α = Radius of 95 per cent circle of confidence (deg) (Fisher 1953).
K = Fisher's precision factor.

Consideration of all the data for rocks likely to possess a stable magnetization and whose ages are known reveals three interesting facts. First, all rocks of Tertiary

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and Quaternary age have directions of magnetization nearly parallel or anti-parallel to the present dipole field. These are shown in Figures 6c and 6d where all the results are plotted as normal directions. Secondly, the mean directions of magnetization of Pre-Tertiary (Mesozoic and late Palaeozoic) rocks from the north-eastern part of Japan are found to be north-westerly (Figure 6a) and thirdly, the mean direction for rocks of late Palaeozoic and early Mesozoic age from the south-west part of Japan are found to be north-easterly (Figure 6b). The mean directions of magnetization are summarized in Figure 6e. It is seen that whilst the mean directions of magnetization of rocks of Tertiary and Quaternary ages from both north-east and south-west Japan are close to the axis of the present dipole field, those of pre-Tertiary age from north-east Japan differ in declination from those in south-west Japan by about 40°.

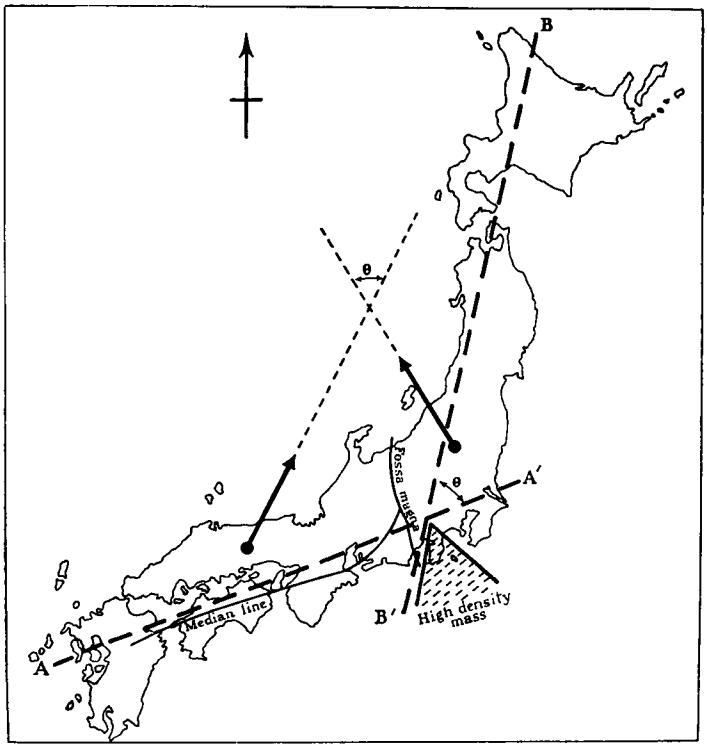


FIG. 7.—Map of the Japanese Islands. AA′ and BB′ are the axes of the south-west and north-east parts of Japan and the arrows represent the mean directions of n.r.m. of Pre-Tertiary rocks sampled from these two regions.

3. Interpretation of results

It is suggested that the difference in declination of 40° for the n.r.m. of rocks collected from south-west and north-east Japan is due to the deformation of the Japanese Islands. Figure 7 shows the position of a zone of plastic deformation known in Japan as the "fossa magna". There are striking differences in geological structure on either side of the "fossa magna" and earthquakes and volcanism are associated with this zone.

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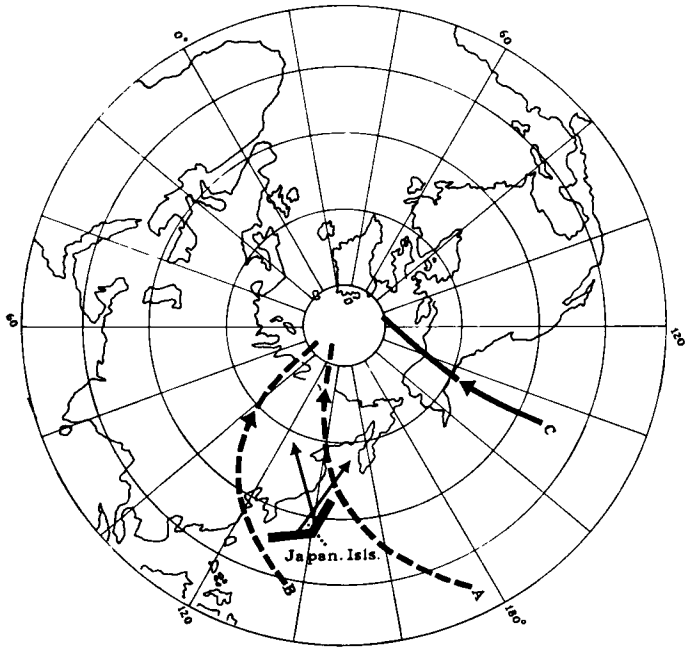


FIG. 8.—Schematic diagram of the bending of the Japanese Islands in relation to the hypothesis of polar wandering. (A) Polar wander path as deduced for North America (B) Polar wander curve as deduced for Europe and (C) polar wander curve as deduced for Japan.

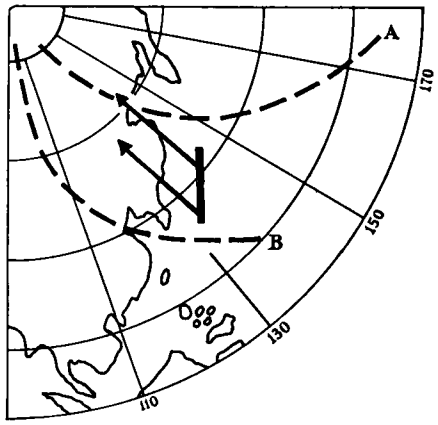


FIG. 9.—A possible interpretation of the direction of magnetization in Pre-Tertiary times by correcting for the bending of the Japanese Islands.

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Tamaki (1955) in an analysis of earthquake data has shown the existence of high density material near the Izu Peninsula, that is, near the "fossa magna". This high density sector may be considered to be a mass of basic rocks intruded into the fracture zones which resulted from the bending of the Japanese Islands along this line. Further, Ehara (1961) found gabbro and granodiorite associated with the andesitic magma extruded at Oshima Island confirming the existence of high density material. The region is also associated with positive gravity anomalies (Matsuyama 1936, and Kumagai 1940 and 1953).

Two lines, AA′ and BB′ in Figure 7 represent the mean axes of south-west and north-east Japan respectively. The angle between these lines (θ) is approximately equal to the difference in the mean directions of magnetization for rocks of pre-Tertiary age collected from these two parts of Japan. It is suggested that the difference in direction of magnetization for rocks from north-east and south-west of Japan is due to the bending of the Japanese Islands along the line of the "fossa magna" and in agreement with the geological evidence. Further, it is possible to infer from the rock magnetism data that the bending took place in early Tertiary times.

4. Implications for studies of polar wandering

In view of this rotational movement of the different parts of the Japanese Islands, it is very difficult to infer the position of the geomagnetic pole with respect to Japan for pre-Tertiary times. If the amount of polar wander between the Mesozoic and early Tertiary is assumed to be small, it is possible to infer that the amount of relative rotation of north-east and south-west Japan is of the order of 40°.

The polar wander curve for Japan as deduced by Nagata & others (1959) differs greatly from those deduced for Europe and North America by Creer & others (1954) and by Runcorn (1959) as shown in Figure 8. The discrepancy might be considerably reduced if the bending of the Japanese Islands is taken into consideration. Figure 9 shows one possible interpretation of removing the effect of the bending of the Japanese Islands and the difficulty of defining the polar wander curve for Japan in pre-Tertiary times.

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